

1. What is image classification? How does it differ from object detection and image segmentation?

Image classification assigns a class label to an entire image. Object detection identifies and localizes objects using bounding boxes, while image segmentation classifies individual pixels to create detailed object regions.

2. Explain the architecture and working principle of a Convolutional Neural Network (CNN).

A CNN consists of convolutional layers, pooling layers, activation functions, fully connected layers, and an output layer. The convolutional layers extract features from images, pooling layers reduce the spatial dimensions, and fully connected layers use the extracted features for classification. The output layer produces the final class prediction.

3. What is the role of Convolutional Layers, Pooling Layers, and Fully Connected Layers in a CNN?

- **Convolutional Layers:** Extract important features such as edges, textures, and shapes.
- **Pooling Layers:** Reduce the size of feature maps and computational complexity while retaining important information.
- **Fully Connected Layers:** Combine extracted features and perform the final classification.

4. Why is image normalization performed before training a deep learning model?

Image normalization scales pixel values to a suitable range, making training more stable and efficient. It helps the neural network converge faster and improves overall training performance.

5. Explain the purpose of activation functions such as ReLU and Softmax in CNNs.

ReLU (Rectified Linear Unit) introduces non-linearity into the network and helps the model learn complex patterns.

Softmax converts the output values into probabilities for different classes, allowing the class with the highest probability to be selected as the prediction.

6. What is a Confusion Matrix? How is it used to evaluate classification performance?

A Confusion Matrix is a table that compares the actual class labels with the predicted class labels. It shows correct and incorrect predictions for each class and helps identify misclassifications. It is also used to calculate Accuracy, Precision, Recall, and F1-Score.

7. Differentiate between Accuracy, Precision, Recall, and F1-Score with suitable examples.

- **Accuracy:** Percentage of total predictions that are correct.
- **Precision:** Percentage of predicted positive samples that are actually positive.
- **Recall:** Percentage of actual positive samples that are correctly identified.
- **F1-Score:** Harmonic mean of Precision and Recall.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = 2 \frac{Precision \times Recall}{Precision + Recall}$$

8. Compare traditional feature-based image classification methods (e.g., SIFT/HOG + SVM) with CNN-based classification.

Traditional methods such as SIFT/HOG + SVM require manually designed features before classification. CNNs automatically learn hierarchical features directly from images.

LAB 10 NITIN

CNN-based methods generally require more computational resources and training data but can learn complex image features automatically. Traditional methods can be simpler when suitable handcrafted features are available.

9. Mention five real-world applications of CNN-based image classification.

1. Handwritten digit recognition
2. Facial recognition
3. Biometric authentication
4. Medical image diagnosis
5. Intelligent document processing

10. How can techniques such as Data Augmentation, Transfer Learning, and Hyperparameter Tuning improve the performance of deep learning-based image classification models?

- **Data Augmentation:** Creates variations of training images through transformations, helping the model generalize better.
- **Transfer Learning:** Uses a model trained on a large dataset and adapts it to a new classification task, reducing training requirements.
- **Hyperparameter Tuning:** Optimizes parameters such as learning rate, batch size, and number of epochs to improve model performance.